

README

HelixCode Integrations

HelixCode supports multiple notification and integration channels to keep you informed about your development workflows.

Supported Integrations

Communication Platforms

1. Slack

Real-time team notifications via webhooks.

Features: - Webhook integration - Custom channels and usernames - Type-based icon emojis - Rich text formatting

[Setup Guide](#)

2. Telegram

Bot-powered notifications for individuals and teams.

Features: - Telegram Bot API integration - HTML message formatting - Personal chats, groups, and channels - Automatic metadata display

[Setup Guide](#)

3. Email (SMTP)

Enterprise-grade email notifications.

Features: - Support for Gmail, Office 365, custom SMTP - TLS/SSL encryption - Multiple recipients - Configurable from addresses

[Setup Guide](#)

4. Discord

Gaming-first webhook notifications.

Features: - Webhook integration - Markdown formatting - Server and channel support - Built-in rate limiting

Setup: Similar to Slack (webhook-based)

Quick Start

1. **Choose your integration(s)** - Select one or more notification channels
2. **Follow the setup guide** - Each integration has a detailed setup guide
3. **Configure HelixCode** - Update your `.env` or `config.yaml`
4. **Test the integration** - Send a test notification
5. **Configure rules** - Define when to send notifications

Configuration Overview

Environment Variables

Slack

HELIX_SLACK_WEBHOOK_URL=https://hooks.slack.com/services/...

Telegram

HELIX_TELEGRAM_BOT_TOKEN=123456789:ABC...

HELIX_TELEGRAM_CHAT_ID=123456789

Email

HELIX_EMAIL_SMTP_SERVER=smtp.gmail.com

HELIX_EMAIL_SMTP_PORT=587

HELIX_EMAIL_USERNAME=your-email@gmail.com

HELIX_EMAIL_PASSWORD=app-password

HELIX_EMAIL_FROM=your-email@gmail.com

Discord

HELIX_DISCORD_WEBHOOK_URL=https://discord.com/api/webhooks/...

config.yaml

```
notifications:
  enabled: true
```

```
rules:
  - name: "Critical Failures"
    condition: "type==error"
    channels: ["slack", "telegram", "email"]
    priority: urgent
    enabled: true

channels:
  slack:
    enabled: true
    webhook_url: "${HELIX_SLACK_WEBHOOK_URL}"
    channel: "#helix-notifications"

  telegram:
    enabled: true
    bot_token: "${HELIX_TELEGRAM_BOT_TOKEN}"
    chat_id: "${HELIX_TELEGRAM_CHAT_ID}"

  email:
    enabled: true
    smtp:
      server: "${HELIX_EMAIL_SMTP_SERVER}"
      port: 587
      username: "${HELIX_EMAIL_USERNAME}"
      password: "${HELIX_EMAIL_PASSWORD}"
```

Notification Types

HelixCode supports different notification types with appropriate visual indicators:

Type	Description	Use Case
info	General information	Status updates, informational messages
success	Successful operations	Task completed, workflow succeeded
warning	Non-critical issues	Performance degradation, warnings
error	Failures and errors	Task failed, execution error
alert	Critical system alerts	System down, critical failure

Notification Rules

Define when to send notifications based on:

- **Event type** - Task failure, workflow completion, etc.
- **Priority level** - Low, medium, high, urgent
- **Metadata conditions** - Custom conditions based on event data

Example rules:

```
notifications:
  rules:
    # Send critical errors to all channels
    - name: "Critical Errors"
      condition: "type==error AND priority==urgent"
      channels: ["slack", "telegram", "email"]
      priority: urgent
      enabled: true

    # Send successes only to Slack
    - name: "Success Notifications"
      condition: "type==success"
      channels: ["slack"]
      priority: low
      enabled: true

    # Send all alerts to Telegram and Email
    - name: "System Alerts"
      condition: "type==alert"
      channels: ["telegram", "email"]
      priority: high
      enabled: true
```

Testing Integrations

Using CLI (if available)

```
helix notify test --channel slack --message "Test notification"
helix notify test --channel telegram --message "Test
notification"
helix notify test --channel email --message "Test notification"
```

Using API

```
curl -X POST http://localhost:8080/api/v1/notifications/test \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -d '{
    "channel": "slack",
    "title": "Test Notification",
    "message": "Testing integration",
    "type": "info"
  }'
```

Using Go Code

```
import "dev.helix.code/internal/notification"

engine := notification.NewNotificationEngine()

// Register channels
slackChannel := notification.NewSlackChannel(webhookURL,
    "#channel", "bot")
telegramChannel := notification.NewTelegramChannel(botToken,
    chatID)
emailChannel := notification.NewEmailChannel(server, port, user,
    pass, from)

engine.RegisterChannel(slackChannel)
engine.RegisterChannel(telegramChannel)
engine.RegisterChannel(emailChannel)

// Send notification
notif := &notification.Notification{
    Title:    "Test",
    Message:  "Testing integrations",
    Type:     notification.NotificationTypeInfo,
}
err := engine.SendDirect(ctx, notif, []string{"slack",
    "telegram", "email"})
```

Roadmap

Planned Integrations

High Priority: - Generic Webhooks (custom integrations) - Microsoft Teams (Workflows API)

Medium Priority: - PagerDuty (incident management) - Jira (issue tracking) - GitHub Issues (repository integration)

Future Considerations: - Mattermost (self-hosted Slack alternative) - Firebase Cloud Messaging (mobile notifications) - Twilio SMS (critical alerts)

See the [Integration Report](#) for detailed implementation plans.

Support

For help with integrations:

1. **Check setup guides** - Each integration has detailed instructions
2. **Review troubleshooting sections** - Common issues and solutions
3. **Check logs** - HelixCode logs contain detailed error messages
4. **Open an issue** - Report bugs or request help on GitHub

Contributing

Want to add a new integration?

1. Implement the `NotificationChannel` interface
2. Add comprehensive tests (unit + integration)
3. Create a setup guide
4. Update this README
5. Submit a pull request

See the implementation report for architecture guidelines.